

# Spectral-front identities for periodic Navier–Stokes and a conditional obstruction to uniform pointwise Osgood closure

ns-singularity-certificate-lab\*

August 2026 — draft

## Abstract

For strong solutions of the unforced incompressible Navier–Stokes equations on  $\mathbb{T}^3$  we prove that the spectral bandwidth  $N_0^2 = \|\nabla u\|_2^2/\|u\|_2^2$  obeys the differential inequality  $\frac{d}{dt} \log N_0^2 \leq KD/(2\nu)$ , where  $D = \|\nabla u\|_2^2$  and  $K = \|\mathbb{P}(u \cdot \nabla u)\|_2^2/\|\nabla u\|_2^4$ , with an explicit nonnegative defect; consequently if  $\int_0^{T_{\max}} KD \, dt < \infty$  the solution is global, and in particular any finite-time breakdown forces the action to diverge. The action admits the exact representation  $\int_0^{T'} KD \, dt = \int_0^{T'} \|\partial_t u\|_2^2/D \, dt + \nu^2 \int_0^{T'} \|\Delta u\|_2^2/D \, dt + \nu \log(D(T')/D(0))$ ; by AM–GM this yields the one-sided criterion  $\int KD \leq 2 \int (\|\partial_t u\|_2^2/D + \nu^2 \|\Delta u\|_2^2/D)$  (the converse comparison fails for explicit decaying solutions), and  $KD$  is dominated pointwise by both the Serrin  $(\infty, 2)$  and the critical  $\|\nabla u\|_{L^3}^2$  actions. On the obstruction side we construct an explicit family of divergence-free trigonometric fields with exact closed-form spectral laws which, conditionally on a single lattice-sum lower bound (open; verified in exact rational arithmetic at small band sizes and numerically to  $N = 32$ ), shows that no Osgood-admissible pointwise field inequality  $K \leq \Phi(\log N_0^2)$  can hold uniformly: the dichotomy cannot be upgraded to unconditional regularity through that route. Proofs are elementary and self-contained modulo classical local theory; all numerical statements are confined to a certificate appendix and support only the conditional hypothesis. The claimed novelty is the exact packaging — the modal identities, their machine-checkable defect ledger, and the certified obstruction family — and overlap with known refinements of the Serrin criteria is possible. Internal review was performed by AI agents and is not human peer review.

## 1 Introduction

The regularity theory of the three-dimensional Navier–Stokes equations is governed by critical quantities whose a-priori control is unknown. We isolate a scalar action,  $\int KD \, dt$ , that (i) bounds the growth of the spectral bandwidth through an exact modal identity with computable defects, (ii) is dominated by the classical critical actions, so its finiteness is a weaker hypothesis than theirs, and (iii) possesses an exact dynamic representation splitting it into a supercritical part (the  $\dot{H}^1$ -bandwidth action) and a harmless logarithm. We then show, conditionally on one lattice estimate that is finitely certified in exact arithmetic and asymptotically open, that the hypothesis

---

\*Machine-assisted research repository; all exact-arithmetic certificates cited below are reproducible from the repository at the commit recorded in the appendix. This draft claims no resolution of any Clay Millennium problem, in either direction.

cannot be verified by any pointwise Osgood-type field inequality: an explicit coherent critical-spectrum family saturates the Bernstein ratio two-sidedly and forces any uniform majorant  $\Phi(\log N_0^2)$  of  $K$  to be of exponential class. The pair of results freezes this lane: the dichotomy is at least as strong as the classical criteria it refines, and the pointwise-Osgood route to making it unconditional is closed modulo that one estimate.

We emphasise what is *not* claimed: no global regularity for all data, no blow-up, no Clay statement. Numerics never enter any proof; they appear only as reproducible exact-arithmetic certificates for the one conditional hypothesis.

## 2 Setting and main results

Let  $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$  with normalised measure,  $u(x, t) = \sum_{k \in \mathbb{Z}^3 \setminus \{0\}} \hat{u}_k(t) e^{ik \cdot x}$  real, zero-mean, divergence-free ( $k \cdot \hat{u}_k = 0$ ), solving  $\partial_t u + \mathbb{P}(u \cdot \nabla u) = \nu \Delta u$  with  $\nu > 0$  and  $u_0 \in H_\sigma^m$ ,  $m > 5/2$ ,  $u_0 \neq 0$ ; let  $u$  be the maximal strong solution on  $[0, T_{\max})$ . Write  $H_r = \sum_k |k|^{2r} |\hat{u}_k|^2$ ,  $D = H_1$ ,  $N_0^2 = H_1/H_0$ ,  $N_1^2 = H_2/H_1$ ,  $z = \log N_0^2 (\geq 0)$ ,  $\mathcal{N} = -\mathbb{P}(u \cdot \nabla u)$ , and  $K = \|\mathcal{N}\|_2^2/D^2$ .

**Theorem 1.** *With the notation above:*

(a)  $\Lambda(t) = \log N_0^2(t) - \frac{1}{2\nu} \int_0^t KD \, ds$  is non-increasing on  $[0, T_{\max})$ ; quantitatively  $\frac{d}{dt} \log N_0^2 = \frac{2\text{Cov}}{N_0^2} - \frac{2\nu V}{N_0^2} \leq \frac{KD}{2\nu}$  with  $\text{Cov} = T_1/H_0$ ,  $V = \text{Var}_p(|k|^2)$ ,  $p(k) = |\hat{u}_k|^2/H_0$ , and an explicit nonnegative two-part defect.

(b) Exactly one of:  $\int_0^{T_{\max}} KD \, dt < \infty$ , and then  $z$  is bounded,  $u \in L^\infty(0, T_{\max}; H^1)$  and  $T_{\max} = \infty$ ; or  $\int_0^{T_{\max}} KD \, dt = \infty$ . In particular  $T_{\max} < \infty \Rightarrow \int_0^{T_{\max}} KD \, dt = \infty$ ; no claim is made that a global solution has finite action.

(c) For every  $T' < T_{\max}$ ,

$$\int_0^{T'} KD \, dt = \int_0^{T'} \frac{\|\partial_t u\|_2^2}{D} \, dt + \nu^2 \int_0^{T'} N_1^2 \, dt + \nu \log \frac{D(T')}{D(0)},$$

and by AM–GM  $\int_0^{T'} KD \, dt \leq 2 \int_0^{T'} (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) \, dt$ ; the reverse comparison fails ( $u = e^{-\nu t}(0, 0, \cos x_1)$  has  $\int KD = 0$  with divergent bandwidth action).

(d) Pointwise,  $KD \leq \|u\|_{L^\infty}^2$  and  $KD \leq C_S^2 \|\nabla u\|_{L^3}^2$ .

(e) If moreover  $KD \leq \Phi(z)D + R$  a.e. with  $\Phi > 0$  nondecreasing,  $\int^\infty ds/\Phi = \infty$ ,  $R \geq 0$ ,  $\int_0^{T_{\max}} R \, dt < \infty$ , then the first alternative of (b) holds.

The proof (Section 3) is a chain of eight elementary lemmas; the only external inputs are the classical local theory, the subcritical Serrin class  $L^\infty L^6$ , and the mean-zero Sobolev embedding.

On the obstruction side, fix  $v_0 \in \mathbb{Z}^3 \setminus \{0\}$  and let

$$\hat{u}_N(k) = \frac{P_k v_0}{|k|^2}, \quad 1 \leq |k| \leq N, \quad P_k = I - \frac{k \otimes k}{|k|^2}. \quad (1)$$

**Lemma 1** (exact family laws).  $u_N$  is real and divergence-free, and with  $S_N = \sum_{1 \leq |k| \leq N} |k|^{-2}$ ,  $T_N = \sum_{1 \leq |k| \leq N} |k|^{-4}$ :  $H_0 = \frac{2}{3} \|v_0\|^2 T_N$ ,  $H_1 = \frac{2}{3} \|v_0\|^2 S_N$ ,  $N_0^2 = S_N/T_N$ ,  $u_N(0) = \frac{2}{3} S_N v_0$ , and  $\frac{2}{3} S_N \leq \|u_N\|_\infty^2 / \|\nabla u_N\|_2^2 \leq S_N$ . Moreover  $N/250 \leq S_N \leq 432N$  for  $N \geq 8$  (crude explicit constants, proven; numerically  $S_N = 4\pi N - 8.7 \pm 0.9$ ) and  $T_N \nearrow T_\infty < \infty$ , so  $N_0^2 \asymp N$ .

**Hypothesis 1** (capacity; open, certified finitely). There is  $c_0 > 0$  with  $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3$  for all large  $N$ .

**Proposition 1** (no Osgood-admissible uniform majorant). *Assume Hypothesis 1. If  $\Phi$  is non-decreasing and  $K(u) \leq \Phi(\log N_0^2(u))$  for every zero-mean divergence-free real trigonometric field  $u$ , then  $\Phi(s) \geq ce^s$  for all large  $s$ ; in particular  $\int^\infty ds/\Phi < \infty$ , so clause (e) of Theorem 1 cannot be activated by any uniform pointwise field inequality.*

**Remark 1.** *Hypothesis 1 reduces, via the proven projection identity  $\mathbb{P}((c \cdot \nabla)u) = (c \cdot \nabla)u$  (constant  $c$ ) and the exact bound  $\|(v_0 \cdot \nabla)u_N\|_2^2 \leq \frac{2}{3}\|v_0\|^4 S_N$ , to the single lattice-sum estimate  $Q(v_0) := \langle (u_N \cdot \nabla)u_N, (v_0 \cdot \nabla)u_N \rangle \geq cN^2$ . The appendix certifies: exact rational  $K(u_N)$  at  $N = 4, 6, 8$  (0.7884..., 1.4345..., 2.0372...; float agreement  $\leq 1.3 \times 10^{-15}$ );  $Q(v_0) > 0$  exactly at those  $N$ ; and that the pairing bound captures 91.2–92.1% of  $\|\mathbb{P}(u \cdot \nabla u)\|_2$  for  $4 \leq N \leq 32$ , essentially  $N$ -independently.*

### 3 Proofs

The full lemma chain L0–L11 with complete proofs is reproduced from the repository file `complete_proof.md`; we give it here in compressed form.

**L0a (regularity of moments).**  $H_0, H_1, H_2$  are continuous;  $H_0, H_1 \in C^1$  with  $\dot{H}_0 = 2\langle u, \partial_t u \rangle$ ,  $\dot{H}_1 = 2\langle -\Delta u, \partial_t u \rangle$  (difference quotients in  $C^1 H^{m-2}$  paired against  $C(L^2)$ );  $\partial_t u = \mathcal{N} - \nu(-\Delta)u$  in  $L^2$ . Positivity of  $H_0$  on all of  $[0, T_{\max})$  is *not* elementary and is proven as L0b *after* L7, by bootstrap: if  $t_1$  is the first vanishing time, then on  $[0, t_1)$  Lemmas 1–4 and 7 apply,  $KD \leq \|u\|_\infty^2 \leq C_m M^2$  with  $M = \sup_{[0, t_1]} \|u\|_{H^m}$ , so  $N_0^2$  is bounded there,  $\int_0^{t_1} N_0^2 < \infty$ , and the integrated energy identity gives  $\dot{H}_0(t_1) > 0$ , a contradiction.

**L1 (ledger; energy neutrality).**  $\frac{1}{2}\dot{H}_r = T_r - \nu H_{r+1}$  ( $r = 0, 1$ ) with  $T_r = \sum |k|^{2r} \text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle$ ;  $T_0 = -\langle u, (u \cdot \nabla)u \rangle = 0$ ; hence  $\int_0^{T_{\max}} D dt \leq H_0(0)/(2\nu)$ .

**L2 (identity).**  $\frac{d}{dt} \log N_0^2 = 2\text{Cov}/\mu - 2\nu V/\mu$  with  $\mu = N_0^2$ : subtract the logarithmic derivatives and use  $H_0 H_2 - H_1^2 = H_0^2 V$ .

**L3 (closable covariance bound).**  $|\text{Cov}| \leq \sqrt{V\|\mathcal{N}\|_2^2/H_0}$ : modal Cauchy–Schwarz  $|\text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle| \leq |\hat{u}_k| |\hat{\mathcal{N}}_k|$ , then vector Cauchy–Schwarz.

**L4 (square completion; (a)).**  $\sup_{V \geq 0} [\alpha\sqrt{V} - \nu V] = \alpha^2/(4\nu)$  with  $\alpha = \sqrt{\|\mathcal{N}\|_2^2/H_0}$  gives  $\frac{d}{dt} \log N_0^2 \leq \|\mathcal{N}\|_2^2/(2\nu H_1) = KD/(2\nu)$ ; the defect is  $\Gamma^{\text{CS}} + \Gamma^{\text{SC}} \geq 0$  as displayed in the repository file, with rational total.

**L5 (dichotomy; (b)).**  $\int KD < \infty \Rightarrow z \leq z_* \Rightarrow H_1 \leq e^{z_*} H_0(0) \Rightarrow u \in L^\infty L^6$  (mean-zero Sobolev), a subcritical Serrin class  $\Rightarrow$  regularity and extension.

**L6 (action representation; (c)).** Expand  $\|\mathcal{N}\|_2^2 = \|\partial_t u + \nu(-\Delta)u\|_2^2$ , use  $2\nu\langle \partial_t u, -\Delta u \rangle = \nu\dot{D}$ , divide by  $D > 0$ , integrate;  $\int \dot{D}/D = \log(D(T')/D(0))$ .

**L7 (dominations; (d)).**  $\|\mathcal{N}\|_2 \leq \|u\|_\infty \|\nabla u\|_2$  and  $\|\mathcal{N}\|_2 \leq \|u\|_{L^6} \|\nabla u\|_{L^3} \leq C_S \|\nabla u\|_2 \|\nabla u\|_{L^3}$ .

**L8 (Osgood closure; (e)).**  $\Omega(z) = \int_0^z ds/\Phi$ ;  $\frac{d}{dt}\Omega(z) \leq D/(2\nu) + R/(2\nu\Phi(0))$ , both integrable;  $z$  bounded; apply L5.

**L9–L11 and the Proposition.** Band symmetry  $\sum_{B_N} k_i k_j f = \delta_{ij} \frac{1}{3} \sum |k|^2 f$  (sign flips kill  $i \neq j$ ; permutations equalise the diagonal); the family laws of Lemma 1 follow by applying it to  $|P_k v_0|^2 = \|v_0\|^2 - (k \cdot v_0)^2/|k|^2$  and to  $u_N(0) = \sum \hat{u}_N(k)$ ; the crude lattice-sum bounds by dyadic-shell cube counting (upper) and an all-positive sub-box count (lower), with bounded dyadic gaps  $c_- \leq s_{2N} - s_N \leq c_+$  for large  $N$ . Given Hypothesis 1,  $K(u_{N_j}) \geq c_0'' e^{s_{N_j}}$  along  $N_j = 2^j N_0$ ; monotonicity of  $\Phi$  interpolates across the bounded gaps, so  $\Phi(s) \geq c_0'' e^{-c_+} e^s$  for all large  $s$  and  $\int^\infty ds/\Phi < \infty$ .  $\square$

## 4 Position among known criteria; limitations

By (d), the finiteness of  $\int K D$  is implied by the classical Serrin  $(\infty, 2)$  action and by  $\int \|\nabla u\|_{L^3}^2 dt$ ; the criterion (b) is therefore at least as strong as both (we do not claim strictly stronger). By (c) and its AM–GM corollary, the criterion is implied by the a-priori control of  $\int (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt$  — the half-derivative-supercritical  $\dot{H}^1$ -bandwidth action — while the exact identity shows the two differ only by  $\nu \log D$ , which can diverge to  $-\infty$  on decaying solutions; this makes precise both why the criterion is strong and why its unconditional verification is out of reach of the present methods. Proposition 1 closes the pointwise-Osgood route conditionally on one certified lattice estimate. All statements are elementary; the contribution is the exact packaging, the certified obstruction family, and the machine-verifiable defect ledger accompanying (a).

## A Certificate appendix (numerics, separated from all proofs)

Reproducible from the repository (`experiments/run_coherent_family_certificates.py`, `run_osgood_gate.py` JSON with SHA-256 sidecars under `outputs/verification_sprint_v1/osgood_gate/`): exact-Fraction verification of the family laws at  $N = 4, 6, 8$  (residual 0); exact  $K(u_4) = 0.78841070\dots$  (full fraction stored),  $K(u_6) = 1.43447181\dots$ ,  $K(u_8) = 2.03724301\dots$ ; float pipeline agreement  $0.0, 4.6 \times 10^{-16}, 1.3 \times 10^{-15}$ ;  $Q(v_0) > 0$  exact at  $N = 4, 6, 8$  with float continuation  $Q/N^2 = 850.7 \rightarrow 1029.5$  over  $N = 8 \rightarrow 32$ ; captured fraction of  $\|\mathbb{P}(u \cdot \nabla u)\|_2$  by the sweeping pairing:  $0.921/0.917/0.915$  (exact,  $N = 4, 6, 8$ ) and  $0.915 \rightarrow 0.912$  (float,  $N = 8 \rightarrow 32$ );  $K/N_0^2 = 0.259 \rightarrow 0.396$  ( $N = 4 \rightarrow 32$ ); Leray retention  $0.780 \rightarrow 0.867$ . These numbers support Hypothesis 1 only and enter no proof.